

Assignment 3 Writeup

A1

For this section, I first inspected the data using methods like `.head()`, `.shape`, `.info()`, and `.describe()`. I also took a look at which columns contained null values and how many each. For the geometry column, I made the decision to drop all null values. This is because geometry is necessary for businesses to show up on a map. It contains latitude and longitude, and without it, they can't be located at all.

This left the number of employees column with no null values, so I didn't do anything else until later. Later on, I log transformed this data since it was heavily right skewed.

Fee paid had a lot of null values. I later decided to do median imputation since dropping null entries would result in a large loss of data, and median is less sensitive to outliers than mean.

A2

From my notebook:

The KMeans clusters capture distinct areas of Vancouver, such as downtown, Mt. Pleasant, East Van, West side, South Vancouver, etc. We can see the downtown core very saturated with businesses, as it is very dense with points in the visualization (light green). Businesses become more sparse as they move away from the busiest areas of the city (downtown, west end, false creek). This map even allows us to see that businesses can be saturated around certain streets. Main street, Hastings, Kingsway, and W 4th are some of the streets that stand out. Since K-means divides the entire study area into a certain number of groups (I chose 6 here), every business is assigned to one cluster no matter what.

DBSCAN instead looks for dense concentrations of businesses. A very large cluster in the downtown core reaching out to the west side and part of east van has been identified because these businesses are closely packed together. Smaller areas with dense businesses have been identified as their own clusters. There has also been a lot of noise identified, meaning these businesses don't belong to a distinct dense business district (because they aren't close enough to a dense area).

KMeans and DBSCAN both have correctly identified downtown as the densest business area of the city. They've also identified other areas where businesses are dense outside of downtown.

However, they do disagree in many areas. KMeans uses a defined number of clusters so even isolated businesses get sorted into one of the 6 clusters. DBSCAN leaves these isolated businesses as noise (-1). Since DBSCAN groups points based on their proximity, it produces

irregular shaped clusters that can sometimes reflect real business areas better, however in this case I think KMeans does a better job. KMeans makes more equally sized clusters where every point belongs in a cluster.

A3

From my notebook:

The clusters here aren't based on Vancouver geography like the other ones were. They group businesses with similar characteristics based on the features. The hope is that they represent certain types of businesses, like large, established businesses with many employees or smaller businesses with less employees and smaller fees. The clusters shown in my visualization don't have the best separation, as there's some overlapping. This means that there are businesses that overlap in terms of features.

Here, it doesn't matter if a business is super far away from another, all that matters is that they're similar based on the set of features.

The Z scores are a bit confusing, as it looks like the first group has less employees than average and pays the lowest fees. It also has the shortest licence duration. These are likely small businesses. The second group has the least employees, yet pays higher fees than the other groups. It has a medium amount of duration compared to the other groups. Group 3 has the greatest number of employees, greatest fee paid, and longest licence duration. I can assume that this is large, established businesses.

B4

Yes, the grouping definitely seems meaningful and corresponds to what the general public knows about these areas. The largest cluster appears to be right in the downtown core, with clusters getting smaller the more they move away from the busiest part of the city.

The clusters also seem to make sense. It seems like areas approaching downtown, like Granview-Woodland, Mount Pleasant, and West End, are clustered together. One interesting cluster contains neighbourhoods like Kerrisdale, West Point Grey, Arbutus-Ridge, Riley Park, and Kensington-Cedar Cottage. These are areas that I know to have lots of boutique shops and restaurants, so it makes sense they'd be grouped together. Kitsilano, Shaughnessy, and South Cambie also make sense to me as they're largely retail spaces. One cluster I'm confused about is Hastings-Sunrise and Sunset. I'm not sure how these areas relate.

B5

Some clusters are geographically close, and some aren't. One cluster that is spread out contains Dunbar-Southlands, Victoria-Fraserview, Renfrew-Collingwood, and Killarney. This

makes sense, since some faraway areas are similar in terms of business density and type. Often, though, this occurs in close-together neighbourhoods.

In A, clusters were grouped by business characteristics like industry and size while B grouped them based on neighbourhoods and the businesses that make them up. What's consistent is we can see concentrated business areas, but what's different is part B shows that some areas are similar even if they're not geographically close.

The most unexpected grouping to me was Hastings-Sunrise and Sunset. I'm assuming these both have a lot of independent restaurant and food businesses, and are therefore grouped together.